Computational Modeling of an Abrupt Silicon p–n Junction

Python-based educational study | Equilibrium depletion approximation | 300 K

Project objective. Investigate how doping concentration affects built-in potential and depletion width in an idealized silicon p–n junction, and visualize depletion-region charge density and electric field.

1. Model and assumptions

The model describes a one-dimensional, abrupt junction at thermal equilibrium with no applied external bias. It assumes uniform doping on each side, complete dopant ionization, non-degenerate statistics, and the depletion approximation. Intrinsic carrier concentration is fixed at 1×10¹■ cm■³ and silicon relative permittivity at 11.7. The doping sweep uses equal acceptor and donor concentrations from 10¹■ to 10¹■ cm■³.

2. Governing equations

```
Thermal voltage: V_T = k_B T / q

Built-in potential: V_bi = V_T ln(N_A N_D / n_i²)

Total depletion width: W = sqrt[(2 ε_Si / q)(1/N_A + 1/N_D)V_bi]

Charge neutrality: N_A x_p = N_D x_n;  W = x_p + x_n

x_p = W N_D/(N_A + N_D);   x_n = W N_A/(N_A + N_D)

Depletion charge: ρ = −qN_A on p side; ρ = +qN_D on n side; ρ = 0 outside.
```

Doping concentrations are converted from cm■³ to m■³ before use in SI equations. The electric field is calculated from Poisson's equation, dE/dx = ρ/ε_Si, with zero field at the depletion edges.

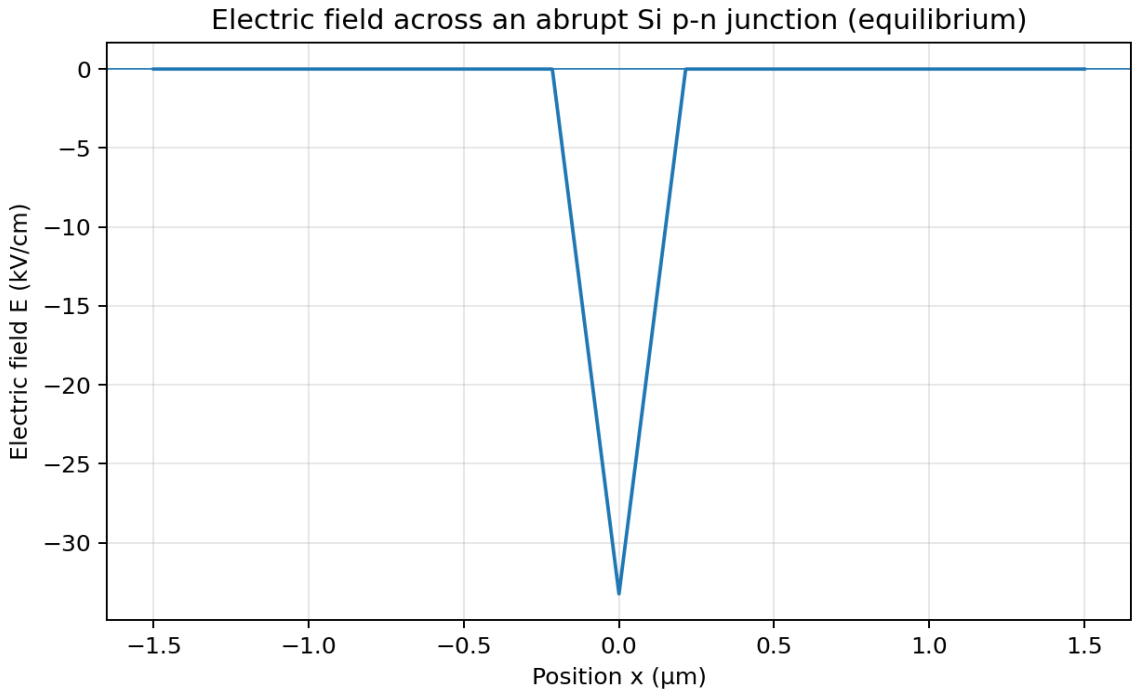
3. Baseline calculation

Parameter	Value
Temperature	300 K
Acceptor concentration N_A	1.0 × 10¹■ cm■³
Donor concentration N_D	1.0 × 10¹■ cm■³
Thermal voltage V_T	0.0259 V
Built-in potential V_bi	0.7143 V
Total depletion width W	0.4298 µm
Depletion into p side x_p	0.2149 µm
Depletion into n side x_n	0.2149 µm

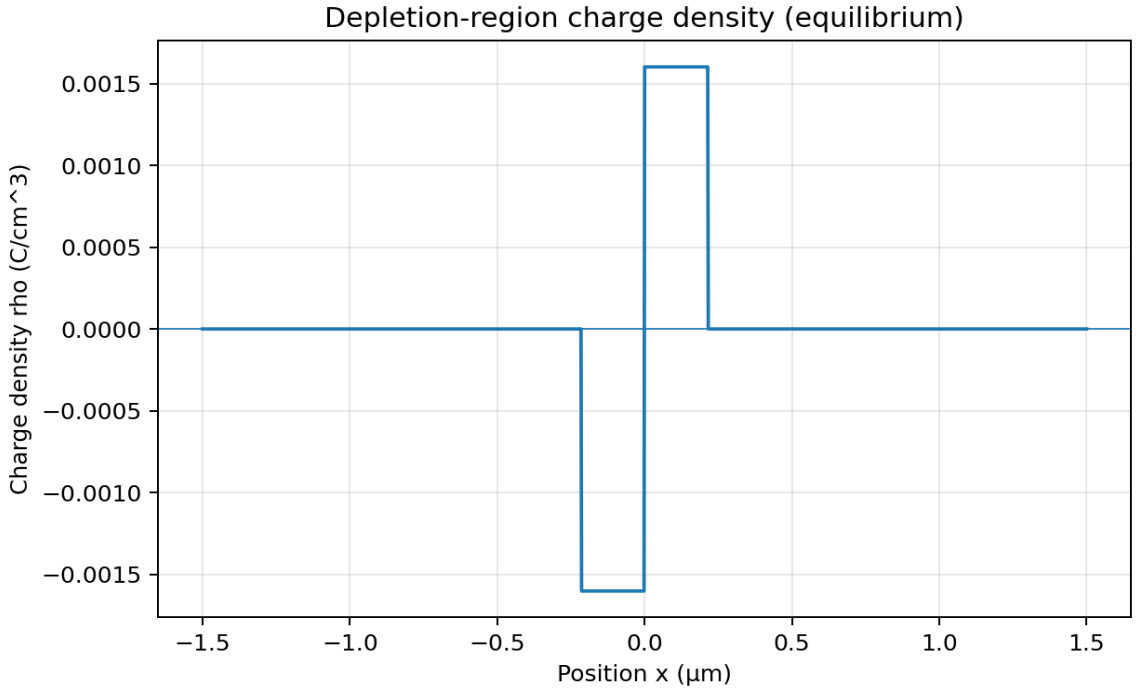
Charge-neutrality check: N_A x_p and N_D x_n are both approximately 2.149 × 10¹■ m■². Since the doping is symmetric, the depletion width divides equally between the two sides.

4. Results

Electric field. The field is zero outside the depletion region, reaches its largest magnitude at the metallurgical junction, and varies linearly within each side under the abrupt-junction depletion approximation.

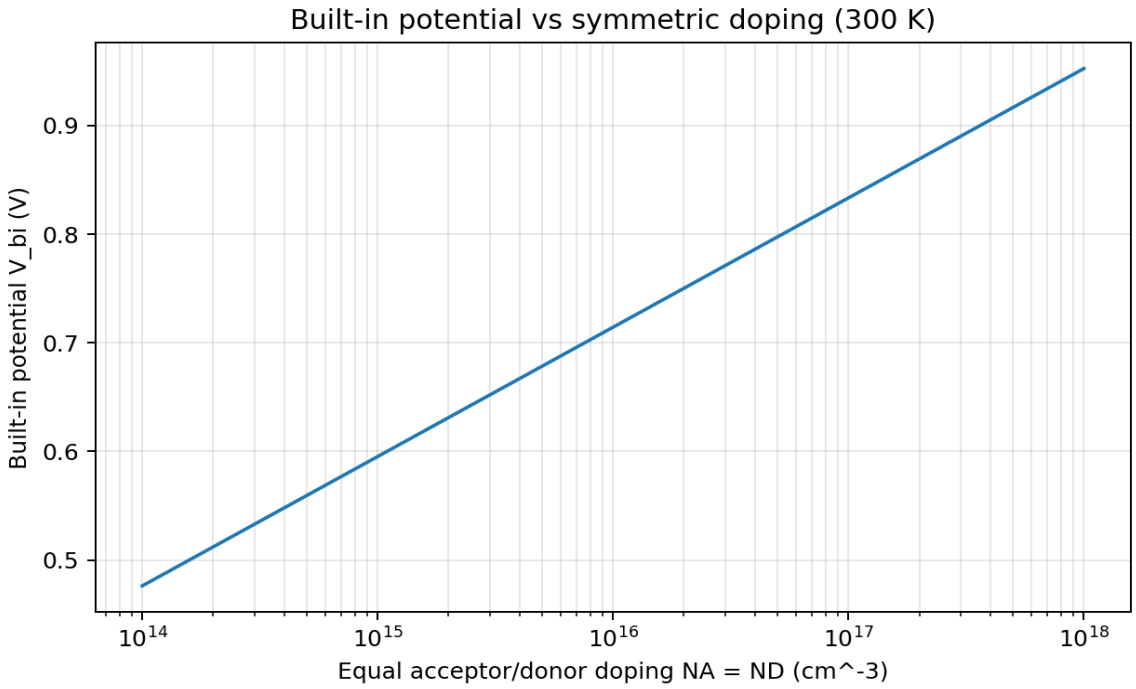


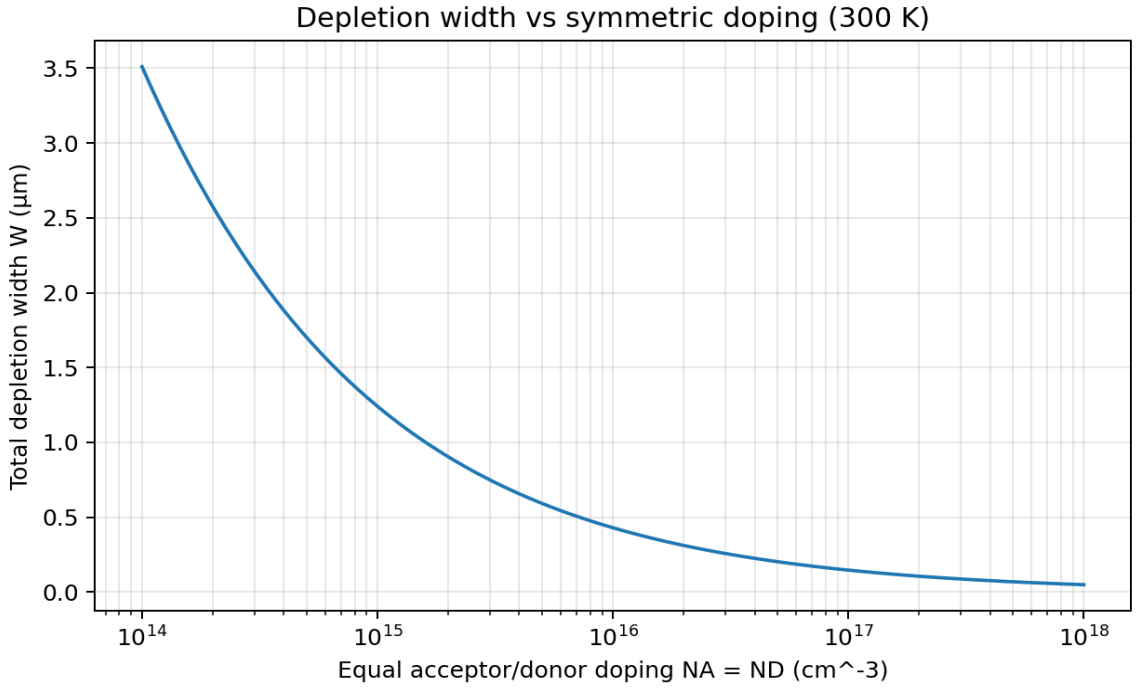
Charge density. Fixed ionized acceptors produce negative charge on the p side; ionized donors produce positive charge on the n side. The idealized depletion-region charge is zero outside these boundaries.



5. Doping sensitivity

For a symmetric junction (N_A = N_D), increasing doping increases the built-in potential logarithmically. The total depletion width decreases because the higher fixed charge density requires a narrower region to support the built-in potential.





6. Validation and limitations

Validation checks implemented in the code include charge neutrality, depletion-width partitioning, and zero electric field at the depletion edges. The field should also be continuous across the junction when there is no sheet charge.

Limitations: This is a simplified analytical/computational model, not TCAD or an experimentally validated device simulation. It does not model current–voltage behavior, carrier transport, recombination, quantum confinement, tunneling, or fabrication effects. The fixed intrinsic carrier concentration and non-degenerate approximation become less reliable at high doping, so

high-doping trends should be interpreted cautiously.

7. Suggested interview explanation

“I implemented a one-dimensional equilibrium model of an abrupt silicon p–n junction in Python. I used the built-in-potential equation and depletion approximation to calculate the depletion width and its division between the p and n sides. I plotted charge density and electric field, then swept equal doping concentrations to examine how doping changes the built-in potential and depletion width. I checked charge neutrality and the depletion-edge boundary conditions. The model is intentionally idealized and does not replace TCAD or include carrier transport.”

Before using this in an application or interview, run the code yourself, understand each equation and unit, independently verify the results with a semiconductor-device textbook or trusted reference, and disclose that it is a simplified model. Cite any external references you use.